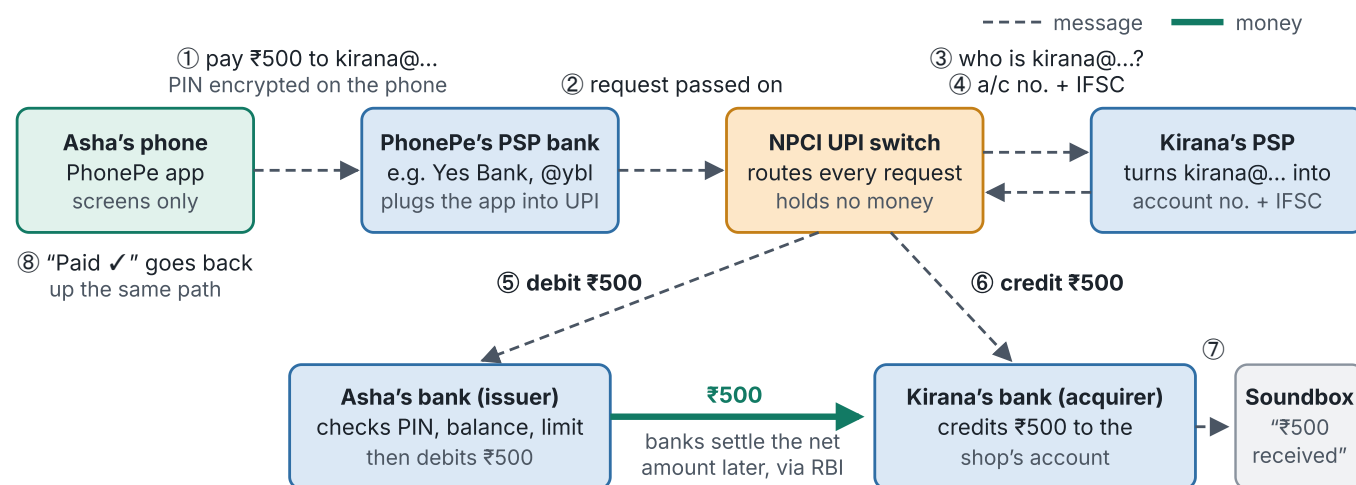


How UPI moves money

7:40 pm, a kirana in Pune. Asha buys ₹500 of groceries, scans the QR code at the counter and types her UPI PIN. Before she has put her phone away, the shop's soundbox says "₹500 received". PhonePe was on her screen the whole time, yet PhonePe never held her ₹500. Here is who did.

24.51 bn UPI payments in Aug 2026, up 22% on Aug 2025	791 mn payments a day, about 9,150 every second on average	₹29.82 lakh Cr moved over UPI in Aug 2026	45.9% · 32.4% PhonePe · Google Pay share of UPI volume; Paytm 8.1%
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Money never sits in PhonePe: the app only carries messages; banks move the money.

Figure 1.1 · One ₹500 UPI payment, step by step. Follow the dashed arrows: four organisations pass messages before any money moves, and the only real money movement (green) is bank to bank. The app you see, PhonePe, owns the screen and the customer relationship, but not the rails. Flow simplified from NPCI's UPI procedural guidelines; PSP handle as used by PhonePe, Sep 2026.

UPI ID (or VPA, virtual payment address) is a nickname for a bank account, such as kirana@ybl. A **TPAP** (third-party app provider) such as PhonePe or Google Pay builds the screens but cannot join UPI alone; it plugs in through one or more **PSP banks** (payment service providers). **NPCI**, owned by a group of banks, runs the switch. The **issuer** is the payer's bank and the **acquirer** is the merchant's.

Who pays for a free payment?

Since January 2020 merchants have paid zero **MDR** on UPI (merchant discount rate: the share of each payment a merchant pays to accept it; roughly 2% on a credit card). Banks and apps carried the cost, and the government paid a small subsidy instead: ₹1,500 Cr in FY25, at 0.15% on small-merchant payments up to ₹2,000. Zero MDR is why a chaiwala takes UPI but not cards. It is also why UPI apps lose money on payments. That changes on **15 October 2026**, when a 0.4% MDR starts on large merchants' payments above ₹2,000.

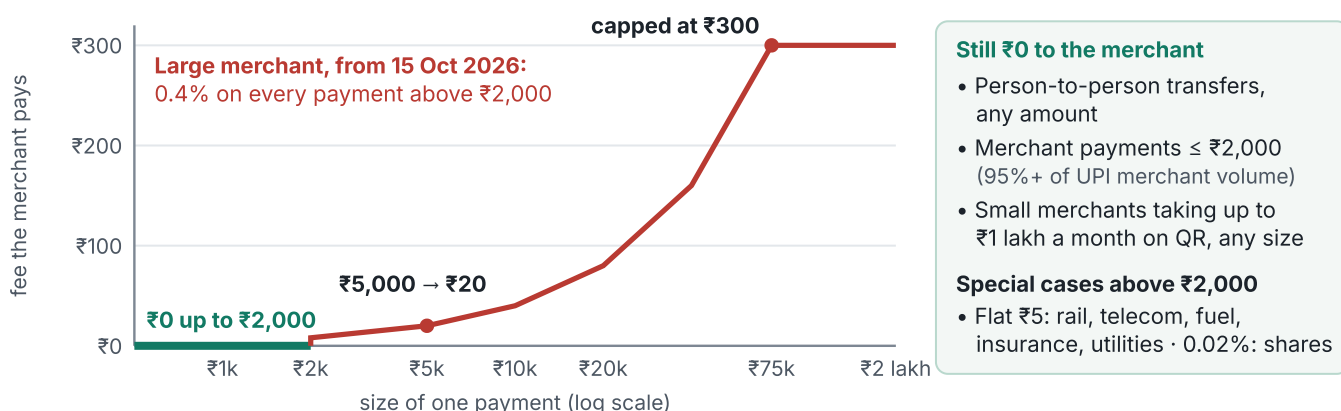


Figure 1.2 · What a merchant pays to accept UPI from 15 Oct 2026. The fee applies only to large merchants and only above ₹2,000, so most everyday payments stay free. How the 0.4% is split between banks, the app and NPCI has not been published, and that split decides whether PhonePe's payments business ever makes money. NPCI MDR FAQ, 15 Sep 2026 (via SCC Online, Business Standard); PIB, FY25 incentive scheme.

Turning points

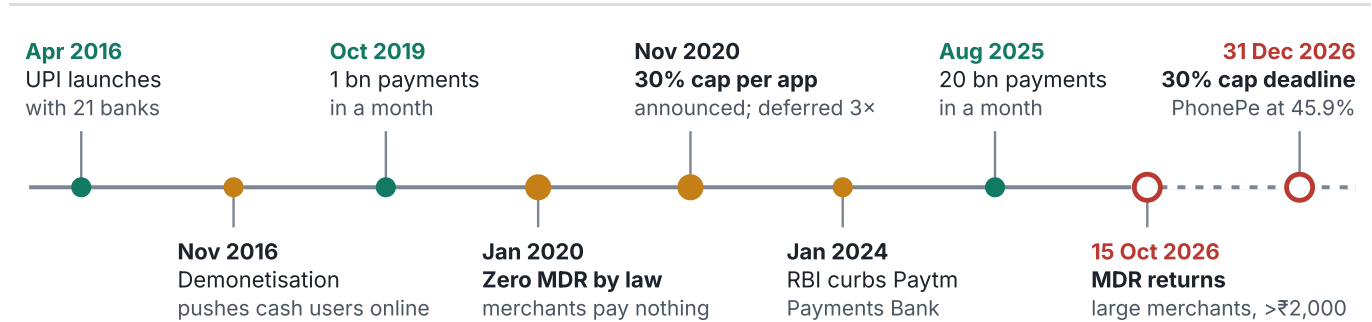


Figure 1.3 · Ten years of UPI in nine moments. Green dots are growth, amber dots are rules, and hollow red dots are still to come. The policy dots shaped the market more than any product launch: zero MDR decided who could make money, and the cap decides who may keep growing. NPCI; *Business Standard* (Aug 2023, Aug 2025); *Business Today* (cap extension, 31 Dec 2024); RBI press release, 31 Jan 2024.

The metrics that matter

Metric	What it means	Worked number, and why it matters
Volume and TPV	Count of payments, and total payment value (₹ moved)	Aug 2026: ₹29.82 lakh Cr ÷ 24.51 bn = ₹1,217 average ticket . A falling ticket means UPI is winning small, everyday payments.
Market share	Share of all UPI payments started in one app, by count	PhonePe 45.9%. The 30% cap is measured on this, over a rolling three months.
Success rate and TD	Share of payments that go through. Technical decline (TD): a failure caused by a bank or system, not the user	Each 0.1 point of TD at 791 mn a day is about 7.9 lakh failed payments daily. Users blame the app they see.
MTU	Monthly transacting users: at least one successful payment in 30 days	Better than downloads or registrations (PhonePe: 70 crore+ registered), because it counts habit.
Take rate	Revenue ÷ TPV	About 0% on most UPI merchant payments today; 0.4% on eligible large-merchant payments from Oct 2026 (₹20 on ₹5,000).

CHECK · "PHONEPE MUST CUT ITS SHARE TO 30% BY JANUARY 2027"

OVERSTATED

The rule, announced in November 2020, says an app above 30% of UPI volume (rolling three months) must stop *onboarding new users*. It does not force existing users off. NPCI has pushed the deadline back three times, most recently to 31 December 2026, and as of September 2026 it is not in force. Watch it: a cap on new users would hand growth to Paytm, Navi and CRED.

KEY IDEA · PAYMENTS ARE THE HOOK

With most UPI payments free to the merchant, apps earn from what the payment habit lets them sell: loans, insurance, broking, bill payments, ads and soundbox rentals. PhonePe's lending arm earned ₹945 Cr in FY26, 2.5× the year before.

IN THE INTERVIEW

Asked "How does PhonePe make money?", avoid "transaction fees". Say: "Most UPI payments earn it close to nothing. The habit is distribution for loans, insurance and bills. From October, large-merchant payments above ₹2,000 add a fee line, but the split isn't public yet."

CASE 1.6 · ANDROID DAU DROPS 40%

Root cause analysis

Medium

IIM A Product Mindset 2026–27, p. 308

Try it first. Read only the question in the blue box. Take 5 minutes and say your structure out loud before you read the candidate's answer.

THE QUESTION

You're the PM for our mobile banking app. Android Daily Active Users (DAU) fell 40% in the last two weeks while iOS DAU stayed steady. Walk me through how you'd diagnose the root cause.

- A** Let me confirm the issue: Android DAU down 40% over two weeks, iOS flat. Correct?
- Q** Yes, that's correct.
- A** A few clarifiers. Is DAU unique users who opened the app at least once that day? Is the drop across both new and existing users? And is it global or localised?
- Q** Yes, unique daily users. Both new and existing are hit. It's global but more pronounced in North America.
- A** First, metric reliability - any change to analytics tooling or event definitions, or any processing delay or mis-aggregation?
- Q** No changes, and no delays or aggregation errors.
- A** Given accurate metrics, external factors: any industry shift hitting Android specifically, or competitor promotions targeting Android users?
- Q** No significant external events or competitor promotions.
- A** Ruling those out, let me segment deeper. Is it consistent across all Android versions, or specific to one? And across all manufacturers, or specific ones?
- Q** It's primarily Android 12 and above, regardless of manufacturer.
- A** Now internal changes - sudden drops usually coincide with a release. Did we ship a new Android version around when this started, with UX, backend, security or permissions changes?
- Q** Yes - a major Android update two weeks ago, specifically for security and permissions. No equivalent iOS update.
- A** That aligns with the timeline. Release quality next: did crash reports or complaints spike post-release, and did the Play Store rating drop?
- Q** Yes - crash reports surged, specifically at app startup, and negative ratings mentioning crashes rose sharply.
- A** Then the crash logs: a consistent error across devices, correlated with the new permissions logic?
- Q** The logs show repeated crashes tied to permission checks during launch, specifically on Android 12+.
- A** Before I conclude, one loose end. If the bug is purely Android-12+ permission logic, the drop should be roughly uniform wherever Android 12+ is common - yet it's more pronounced in North America. The most likely reconciliation is a staged rollout (the update reaching North America first) or higher Android-12+ penetration there. Can we confirm the rollout sequence and the regional OS-version mix?
- Q** Yes - the update was rolled out to North America first.
- A** To summarise the path: metrics verified; external factors and competitors ruled out; segment isolated to Android 12+ globally, surfacing first in North America via a staged rollout; cause narrowed to our recent security/permissions update crashing the app at startup on Android 12+.
- Solutions:**
- Short term:** roll back to the last stable version; if rollback isn't feasible, ship a hot-fix correcting the permissions logic and deploy fast.
- User communication:** tell affected users, give clear update instructions once fixed, and consider a goodwill gesture to rebuild trust.
- Long term:** strengthen QA for Android OS upgrades, permissions and startup paths; add automated alerting on DAU and crash-rate anomalies; and phase future rollouts so a regional canary catches this earlier.
- Q** Agreed. Thank you - we can close the interview.

WHAT WORKED · WHAT WAS MISSING

✓ **Checked the data first.** Asking about tracking changes and pipeline delays is rare: only 1 of the 12 RCA cases in

✗ **Never asked "sudden or slow?"** A sudden drop usually means something broke; a slow one means users are

this book does it.

- ✓ **Pinned the definition** (unique users, at least one open a day) and checked new vs existing users.
- ✓ **Segmented by OS version and maker**, which turned “Android” into “Android 12+”.
- ✓ **Closed the loose end**. The North America skew didn’t fit a pure OS bug, so the candidate proposed a staged rollout and asked for proof. This is the best moment in the case.

drifting away. That one question points straight at the update and saves three turns.

- ✗ **Looked outside first**, although the drop began right after our own update.
- ✗ **Never checked the size**. Does the crash explain all 40%, or is something else going on too?
- ✗ **Forgot the business cost**. In a banking app, users who can’t open it can’t pay, and they call support.

CHECK · “ROLL BACK TO THE LAST STABLE VERSION” **OVERSTATED**

On Android you can’t take an update back from phones that already have it. You can only stop it reaching more phones (the Play Store lets you pause a gradual release) and then send a *newer* version containing the old, working code. Teams that put risky features behind an on/off switch controlled from their own servers (a “feature flag”) can simply switch the feature off. More in M9.

THE SCOPING KIT · FIVE QUESTIONS BEFORE ANY ANSWER

A good doctor doesn’t write a prescription the moment you say “headache”. They ask where it hurts, since when, and whether anything changed: new food, new medicine, less sleep. The scoping kit is your version of those questions. It takes about a minute and stops you solving the wrong problem.

Same questions, everyday setting: planning a friend’s birthday, you’d ask what they’d enjoy (goal), who is coming (who), what’s already booked (today), and the budget and date (limits).

1 What exactly is it? So you and the interviewer mean the same thing. <i>“Android daily users down 40% in two weeks, iPhone steady. Right?”</i> ✓ asked	2 What’s the goal? More users, more money or fewer failures lead to different answers. <i>Already given here: get the users back.</i> not needed	3 Who is affected? New and old users, or two cities, behave differently. <i>“New users, existing users, or both?”</i> ✓ asked	4 What changed? Most problems start with a change. Ask what exists today and why now. <i>“Did we release anything just before?”</i> ✓ asked late	5 What are the limits? Country, platform or deadline shrinks the problem. <i>“All countries, or one?”</i> ✓ asked
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For a “number dropped” case, add two or three of these: What exactly is counted? ✓ · Sudden or slow? ✗ **skipped** · Where is it worst? ✓ · Can we trust the number? ✓ **rare, and impressive**

Rule of thumb: don’t ask everything. Ask 3–4, say one assumption out loud (“I’ll assume this is worldwide”), and begin.

STRONGER ANSWER



Figure 1.4 · A “metric dropped” answer in six moves. Every RCA case in this book can be answered in this order. Each turn below is tagged with its move. *Author’s summary of the RCA sequence.*

A ① CLARIFY

Before I start, two quick questions. Did the drop happen suddenly, on one day, or slowly over the two weeks? And did we change anything just before it began, such as an app update?

Q Suddenly, the day after we released a new Android version.

Why this move: a sudden drop usually means something broke; a slow one means users are drifting away. One answer and the update is the main suspect.

A ② RULE OUT · ③ FIND WHERE

Two quick checks first: is the number real, with no tracking bug, and did anything outside change, like a rival’s offer? If not, can we split Android users by their Android version, and by whether they installed the update?

Q The number is real and nothing outside changed. For users on Android 12 or newer who installed the update, the app crashes the moment they open it.

Why this move: clear the cheap explanations in one breath, then narrow down to exactly who is hit.

A ④ CHECK THE SIZE

Does that explain the whole 40%? Say 70% of our Android users are on Android 12 or newer, and 60% of them have updated (illustrative). $70\% \times 60\% = 42\%$ who can't open the app. That's close to 40%, so this one bug explains the drop. North America saw it first because the update reached it first.

Why this move: if the numbers didn't add up, there would be a second cause still to find.

Q Good. What do you do now, and how will you know it's fixed?

A ⑤ FIX · ⑥ MEASURE

Today: stop the update reaching more phones. Send out a fixed version fast, because on Android you can't take an update back. Text affected users to update, since they can't open the app to read a notice.

Goal (North Star): Android daily users back to normal within a week. **Safety check (guardrail):** logins and payments still work after the rushed fix. **Trade-off:** a rushed fix is less tested, so it goes to 10% of users for a few hours before everyone. **Next time:** test every release on the newest Android versions.

FRAMEWORK · FINDING THE CAUSE: FOUR DOORS, THEN 5 WHYS

1 · The situation. A number has dropped and you have 25 minutes. The usual mistake is to grab the first idea ("a rival must have run an offer") and spend the case defending it.

2 · An everyday example. Your scooter won't start. You don't open up the engine first. You check the easy things first, in an order that works just as well for an app:

Door	The question	Scooter won't start	Android users fell 40%
1 · Data	Is the reading real?	Is the fuel gauge just broken?	Is the tracking broken?
2 · Outside	Did the world change?	Is there a petrol strike?	Did a rival launch an offer?
3 · Inside	Did we change something?	Did the mechanic work on it yesterday?	Did we release an update?
4 · Where	Where exactly is it?	Only on cold mornings?	Only on Android 12 or newer?

3 · The framework. Knock on the doors in order, because the first ones are quickest to check. If a clue points at one door, go there early: a sudden drop right after an update points at door 3. When a door opens, keep asking "why?" (the **5 Whys**) until you reach something a team can fix. That is usually a missing step in a process, not a person to blame.

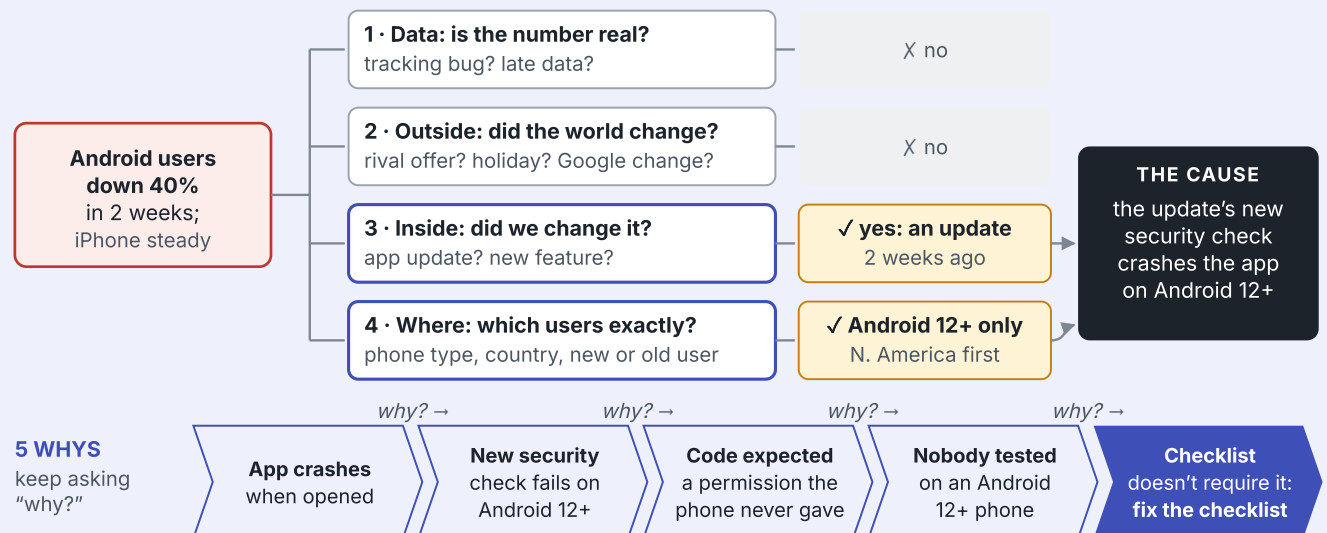


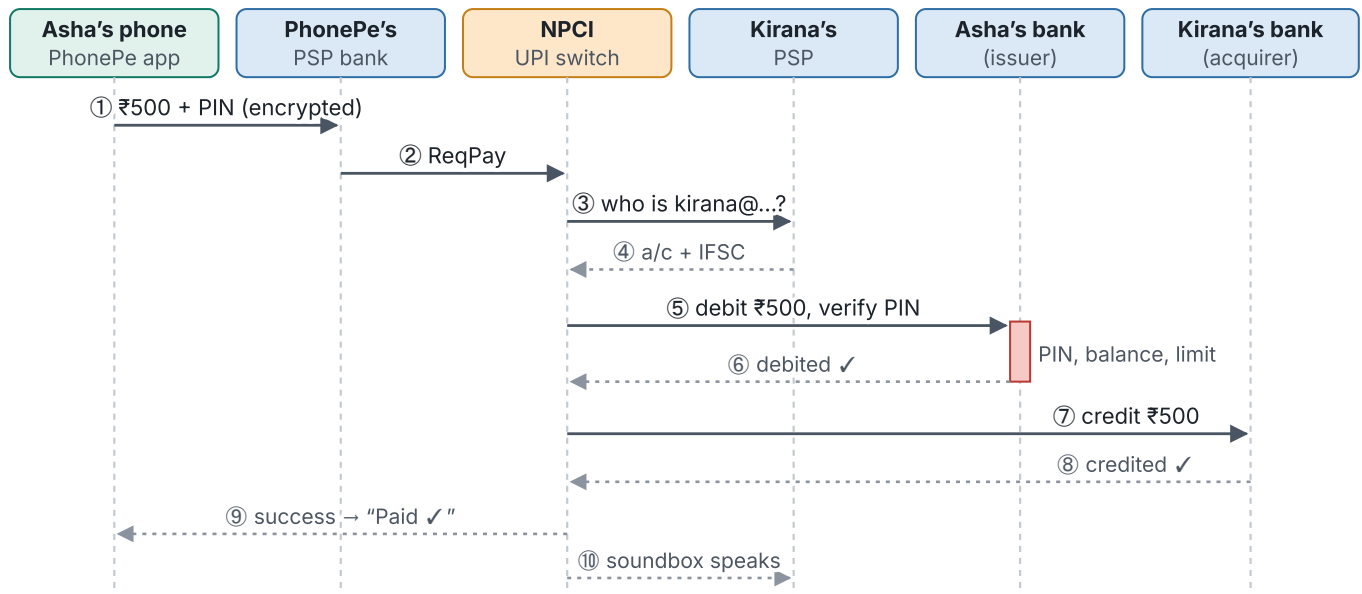
Figure 1.5 · The four doors and the 5 Whys, on this case. Two doors stay shut, two open (blue) and point to one cause. Then "why?" five times walks from the symptom to the real fix: a checklist, not a person. *Case facts from IIM A p. 308; whys 3–5 are illustrative.*

4 · When it misleads. Your scooter has a flat tyre *and* an empty tank. Fix the tyre and it still won't go. So once you find a cause, check it explains the whole drop (move ④ above). 5 Whys also goes wrong if you stop at "the developer made a mistake". People always make mistakes; the fix is the check that should have caught this one.

5 · In the interview. Say it once, in plain words: "First I'll check the number is real, then whether something outside changed, then what we changed, then which users are hit." Then follow the clues, and say out loud when you skip a door and why.

2 seconds inside a UPI payment

Back at the kirana. Between Asha's thumb on "Pay" and the soundbox speaking, about ten messages cross five organisations. Knowing that path lets a PM answer the question every payments team hears weekly: "The money left my account but the shop didn't get it. Where is it?"



Where ~1.9 s goes (illustrative)

NPCI timeout: 15 s for pay requests, since 16 Jun 2025

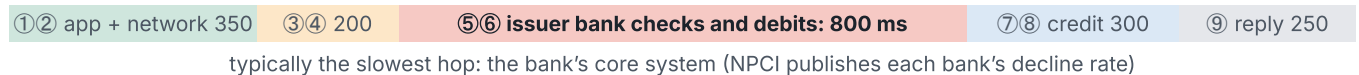


Figure 1.6 · A UPI payment as a sequence of messages (time runs downwards). Solid arrows are requests, dotted arrows are replies. Only step ⑤ touches Asha's money, and it happens deep inside her bank, far from the app she is looking at. Message names from NPCI's UPI specification; timings are illustrative, no public per-hop data exists. Timeout: NPCI circular, 26 Apr 2025.

API: a fixed menu of requests one system accepts from another. In UPI, *ReqPay* means "pay this".

Switch: a router for payment messages. NPCI's switch sees every UPI payment and holds none of the money.

Timeout: how long a system waits before giving up. UPI allows 15 s for pay requests and 10 s for status checks.

Idempotency: sending the same request twice has the effect of sending it once. Each UPI payment carries a unique ID, so a retry can't debit twice.

Pending ("deemed"): debit done, credit not confirmed. A status check or a reversal settles it later.

p95 latency: the time within which 95% of requests finish.

Why the average lies

Take 100 payments: 90 finish in 1.2 s, 8 take 3 s and 2 hang until the 15 s timeout. The mean is $(90 \times 1.2 + 8 \times 3 + 2 \times 15) \div 100 = 1.62$ s, which looks healthy. But the 95th payment in line (**p95**) takes 3 s and the 99th (**p99**) takes 15 s. Users feel their own payment, not the average: the unlucky two stand at the counter staring at a spinner. At 791 mn payments a day, 2 in 100 would be 1.6 crore stuck payments daily (illustrative; real UPI failure rates are lower).



Figure 1.7 · The same 100 payments, three summaries. The mean hides the two payments that hang for 15 s; p95 and p99 show them. Payments teams set targets on p95/p99 and success rate for this reason. Illustrative numbers.

Success rates multiply

A payment succeeds only if every hop succeeds, so the hop success rates multiply. Five hops at 99.9% each give $0.999^5 \approx 99.5\%$: one payment in 200 fails, although no single hop looks bad. Let one issuer bank slip to 99.0% and the chain drops to $0.999^4 \times 0.99 \approx 98.6\%$, or one in 72. That is why payments teams track success rate *per bank*, and why an app's success rate can fall with no change to the app.

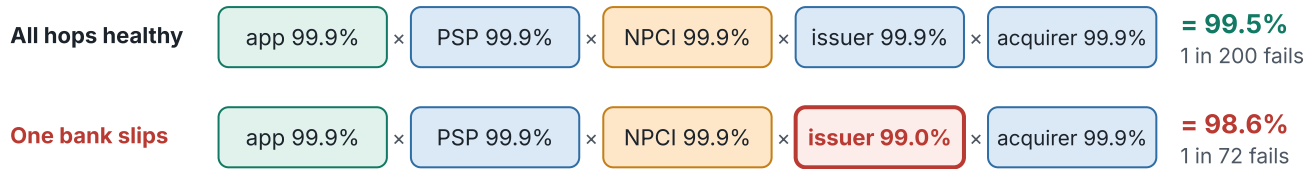


Figure 1.8 • One weak link nearly triples failures. The app, PSP and NPCI did nothing wrong in the second row, yet users see almost three times as many failures and blame the app on their screen. *Illustrative rates; arithmetic exact.*

Uptime	Down per month	Missed*
99.9% (three nines)	43 min	~2.4 crore
99.99%	4.3 min	~24 lakh
99.999%	26 s	~2.4 lakh

*At Aug 2026's average of ~9,150 payments a second (791 mn ÷ 86,400 s). Peaks run higher, so real losses would be larger. Each extra nine cuts downtime tenfold and usually costs far more than ten times as much.

WHEN IT BREAKS • "MONEY GONE, SHOP NOT PAID"

If step ⑤ debits Asha but the reply to ⑦ or ⑧ is lost, the payment is pending. The app must not say "failed" (Asha pays again and is out ₹1,000) or "success" (the kirana hands over goods unpaid). RBI's rule: a failed UPI payment that debited the customer must be reversed by T+1, or the bank pays ₹100 a day.

KEY IDEA • THE PM DECISIONS THIS DRIVES

- What the pending screen says, and how often it re-checks status.
- Retry rules: re-check status freely, never resend a payment with a new ID.
- Routing: PhonePe spreads users across PSP banks (@ybl, @ibl, @axl), so a failing bank can be routed around.
- Dashboards: success rate and p95 per bank, not one average.

IN THE INTERVIEW

"UPI failures are up on our app. Why?" Split by hop before you guess: one issuer bank, one PSP bank, or all of them? Failures clustered on one bank point outside your app. Failures everywhere point at your app, or at NPCI.

PhonePe: free payments, paid loans

Walkthrough (illustrative persona). Meena, 41, runs a tailoring shop in Indore. On the 5th of each month she opens PhonePe, taps *Electricity*, picks her board, and the app fetches her ₹1,240 bill through Bharat BillPay. One PIN later it is paid, and a banner offers her a business loan. That banner is where PhonePe's revenue is heading.

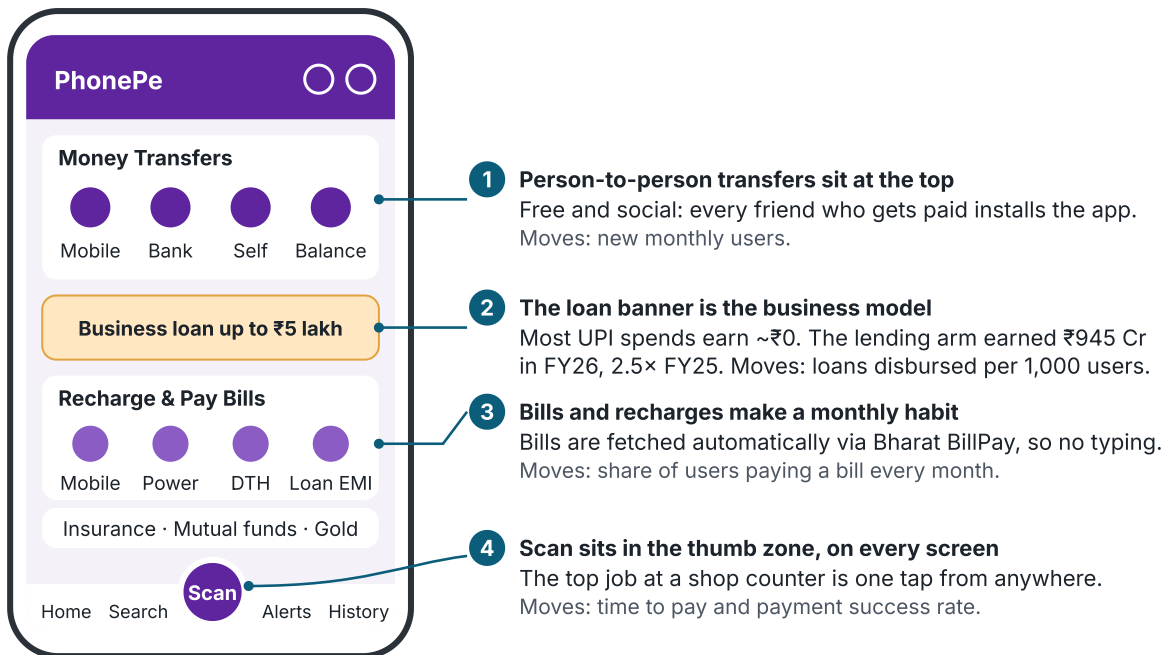


Figure 1.9 • PhonePe's home screen, read as a business plan. Payments (1, 4) bring people in and keep them coming back; the loan banner and financial tiles (2) earn the money. *Simplified redraw, not pixel-accurate; layout varies by app version, Sep 2026. Revenue: company filings via Outlook Business, FY26.*

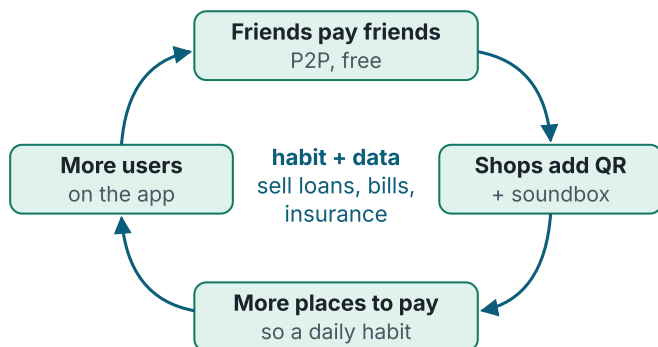


Figure 1.10 • The core loop. Each turn adds users and shops for free; money is made only at the centre, by selling to the habit. *Author's model.*

Operating revenue	₹7,920 Cr
Lending arm	₹945 Cr (2.5×)
Insurance arm	₹191 Cr
Broking arm	₹108 Cr
Govt UPI incentive	₹286 Cr
Net loss	-₹2,792 Cr

Arms are subsidiaries, counted inside the total.

Figure 1.11 • PhonePe FY26 money in and out. Revenue grew 11%; the loss grew 62%, driven by stock-based pay and write-downs. *Inc42, Outlook Business, Entracker (company filings), FY26.*

NORTH STAR Monthly transacting users: at least one successful payment in 30 days.	INPUTS New P2P senders · live merchant QRs · payment success rate · loans per 1,000 users.	GUARDRAILS Fraud loss per ₹1 lakh moved · reversal complaints · loans 30+ days overdue.
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Moat, in plain words. Habit (45.9% of UPI payments in Aug 2026 started on PhonePe) and 5 crore+ merchants with its QR codes and soundboxes. The crack: UPI makes switching free, because any UPI app scans any QR. **2025–26 moves:** IPO papers filed Jan 2026, listing paused Mar 2026; Pincode (quick commerce) shut; Indus Appstore written down; large-merchant MDR from 15 Oct 2026.

What I'd build next

1. A pending payment at a counter breaks trust on both sides → soundbox says “payment pending, PhonePe guarantees it” for pending payments under ₹500 → repeat payments at that shop within 7 days.
2. Small merchants pay no MDR, so they earn PhonePe nothing → use soundbox takings as the credit signal for working-capital loans → approval rate and 30+ day defaults, soundbox merchants vs others.